

P13. A Systematic Survey of Control Techniques and Applications in Connected and Automated Vehicles

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2023 arXiv survey PDF pages: 32 Relevance: 72/100

1. 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

研究背景与痛点 / Background & Pain Points

这篇论文位于“CAV control survey”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of CAV control survey. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Broad CAV control overview; not intersection-smoothing specific.

核心科学问题 / Core Scientific Question

CAV 控制从状态估计、轨迹跟踪到协同控制有哪些主线方法，它们如何支撑交叉口协同研究？

What are the main CAV control families from state estimation to trajectory tracking and cooperative control, and how do they frame intersection research?

科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Systematic survey from vehicle state estimation and trajectory tracking to collaborative CAV control applications.
- 安全/避碰贡献 (safety contribution): Reviews vehicle-control approaches relevant to safety and collaborative control.
- 平滑/停止避免贡献 (smoothing contribution): Frames comfort, energy saving, and transportation efficiency as central CAV control goals.
- 创新力度评判 (reviewer judgement): 综述价值强但原创方法弱：适合作为背景和分类框架引用，不应当作为核心算法依据。

核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Systematic survey of CAV control	CAV 控制技术系统综述	该论文的核心方法族。

教材式学习路径 / Textbook–Style Study Path

- 第一遍先读首页截图、摘要与引言，确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading，对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验，判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式：通用车辆控制闭环 / Generic vehicle–control loop，并解释每个符号的物理意义？
 - 能否画出输入–模型–算法–控制–验证的数据流程图？
 - 能否指出至少两个强假设，并说明它们在真实交通中何时失效？
 - 能否复述实验基准是否公平，指标是否覆盖效率、安全、能耗和计算时间？
 - 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验？
-

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A Systematic Survey of Control Techniques and Applications in Connected and Automated Vehicles

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Abstract—Vehicle control is one of the most critical challenges in autonomous vehicles (AVs) and connected and automated vehicles (CAVs), and it is paramount in vehicle safety, passenger comfort, transportation efficiency, and energy saving. This survey attempts to provide a comprehensive and thorough overview of the current state-of-the-art of vehicle control technology, focusing on the evolution from vehicle state estimation and trajectory tracking control in AVs at the microscopic level to collaborative control in CAVs at the macroscopic level. First, this review starts with vehicle key state estimation, specifically vehicle sideslip angle, which is the most pivotal state for vehicle trajectory control, to discuss representative approaches. Then, we present symbolic vehicle trajectory tracking control approaches for AVs. On top of that, we further review the collaborative control frameworks for CAVs and corresponding applications. Finally, this survey concludes with a discussion of future research directions and the challenges. This survey aims to provide a contextualized and in-depth look at state-of-the-art in vehicle control for AVs and CAVs, identifying critical areas of focus and pointing out the potential areas for further exploration.

Index Terms—Autonomous vehicles; Connected and automated vehicles; State estimation; Vehicle sideslip angle; Trajectory tracking control; Collaborative control

I. INTRODUCTION

Automated driving and collaborative driving automation technologies are revolutionizing future transportation systems regarding reducing traffic congestion, enhancing safety, and improving energy efficiency [1]–[3]. AVs and CAVs, as the key instantiation of automated driving and collaborative driving automation become more and more widely deployed and they benefit both individuals and society [14], [15]. In the past few decades, manufacturers and scholars have been thriving in automotive industrialization to maximize benefits from automated driving and collaborative driving automation [16]–[22]. The hierarchical framework, including perception,

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Typically, the LQR approach is suitable for optimizing linear systems, where the parameters of Q and R need to be tuned to facilitate the system's convergence and stability. For vehicle trajectory tracking control, the LQR method is usually designed based on VDB models. Pao et al. present an LQR optimal trajectory tracking control with a bicycle VDB model, and they optimized the weight matrices of Q and R using a simultaneous perturbation random approximation algorithm [160]. Choi et al. design the trajectory tracking control scheme by winding road disturbance compensator (WRDC) gain on a curved road from the dominant state for the lateral error in the lane-keeping control, and then the WRDC is applied to an LQR trajectory tracking control to compensate for the inaccuracy of linear system. Eventually, compared to the classical LQR methods, the proposed algorithm is demonstrated to be more effective [161]. Especially, vehicle trajectory tracking control under a critical driving situation is essential and should not be ignored in real applications [162]. Menhour et al. employ an LQR method with the LMI framework to solve the problem of switching steering control considering critical driving conditions [163]. In some cases, the kinematic model is also utilized in the LQR method. Akula et al. propose a non-linear kinematic Lyapunov-based control to address the problem of vehicle trajectory tracking control. The corresponding parameters are obtained through the LQR approach with a linear matrix inequality (LMI) expression form [164].

Based on the above survey, the LQR approach can provide an optimal control solution that minimizes the cost function, which can result in superior tracking performance through online or offline optimization. In particular, many researchers use the LQR method with LMI to achieve control robustness via typical Lyapunov function designs. However, LQR control is not suitable for systems with large control inputs because it assumes that the control input is small and does not cause significant deviations from the desired trajectory. Therefore, in some critical driving conditions, such as considerable road curvatures or low road adhesion, the trajectory tracking control performance may be compromised and result in poor robustness. Moreover, although LQR tracking control can handle nonlinear systems by linearizing the dynamics around the desired trajectory, linearizing the highly nonlinear and rapidly time-varying vehicle model is likely to result in model distortion. Therefore, more advanced techniques may need to be explored for the vehicle system.

3) SMC control: To address the issue of linearization assumptions in LQR, SMC is introduced in AVs trajectory tracking control. SMC is a robust control method for nonlinear systems with parametric uncertainties and external disturbances [165]. The challenge in implementing SMC is to design a super-twisting plane to ensure robust stability and reduce the chattering phenomenon. The system function of SMC is as follows:

$$\dot{z} = f(t, x) + g(t, x)u(t) \quad (15)$$

where f and g are continuous functions; x and u are the state vector and the control input, respectively. Then a sliding

variable s with a derivative is expressed as follows:

$$\dot{s}(t, x) = \phi(t, x) + \Phi(t, x)u(t) \quad (16)$$

The sliding mode trajectory tracking control aims for the system to converge on the sliding surface, defined as $s = 0$. To achieve this goal, u_0 , u_{reach} , u_{maint} , and C_0 should satisfy the following constraints:

$$\begin{cases} s \in \mathbb{R}^n, |s(t, x)| < u_0 \\ |u(t)| \leq U_{\text{max}} \\ 0 < h_{\text{reach}} < C_0 |s(t, x)| \leq h_{\text{maint}} \\ | \phi(t, x) | < C_0 \end{cases} \quad (17)$$

Thus, the SMC can be given as follows:

$$\dot{s} = u(t) = u_1 + u_2 \begin{cases} u_1 = -\alpha |s|^\beta \text{sign}(s), \tau \in [0, 0.5] \\ u_2 = -\beta \text{sign}(s) \end{cases} \quad (18)$$

Hence, the super-twisting algorithm based on a sliding surface yields the control signal of the steering angle δ . Where α and β are positive constants. And they should satisfy the following conditions:

$$\begin{cases} \beta \geq \frac{1}{2} \\ \alpha \geq \sqrt{\frac{2\beta U_{\text{max}}}{\Gamma(1-\beta)U_{\text{max}}^{1-\beta}}} \end{cases} \quad (19)$$

For the trajectory tracking control algorithm of AVs, the lateral error is described by $e = y - y_{\text{ref}}$. And the desired lateral acceleration $a_{y_d} = \ddot{y}_{\text{ref}}$ can be expressed as a function of R , which is the radius of the road curvature. According to the VDB model, the error model is expressed as follows:

$$\dot{e} = \dot{y} + \dot{y}\dot{\varphi} - \frac{e^2}{R} \quad (20)$$

Due to its insensitivity to model uncertainties, SMC has become a popular nonlinear method for the trajectory tracking control of AVs, and the VDB model is widely exploited to describe vehicle motion. Akermi et al. employ a fuzzy system to automatically adjust the gain of SMC to compensate for variations in system parameters [166]. Dai et al. propose to integrate SMC with particle swarm optimization (PSO) to improve the trajectory tracking control robustness of AVs. In detail, the authors design an SMC-based steering trajectory tracking control algorithm to eliminate lateral and heading errors [167]. However, an inappropriate sliding surface may cause system oscillations. To eliminate this drawback, Li et al. present an adaptive SMC control scheme to improve handling and stability by reducing jittering. In this method, the weight of the sliding surface is adaptively scheduled according to the stability index [168]. Wang et al. design the high-order sliding mode to effectively reduce system chattering and improve control accuracy [169]. As high-order sliding modes suffer from computation workload, the second-order sliding mode remains the mainstream to ensure the real-time performance of the algorithm. Alternatively, some scholars integrate SMC with other control methods to eliminate the jittering phenomenon. Zhang et al. present an optimal preview linear quadratic regulator (OPLQR) based on the SMC approach with a 2-DOF vehicle VDB model to enhance the smoothness of the vehicle motion after obtaining the desired steering angle [170]. Nasr et al. achieve improved trajectory tracking performance by

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p.1 · full_page

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model to enhance the performance of sideslip angle estimation [127]. Similarly, Novi et al. integrate an artificial neural network, with UKF based on a kinematic model to estimate sideslip angle [128]. Compared with the pure AIA approach, the hybrid approach can improve the interpretability of the estimation method to a certain extent.

The survey mentioned above highlights that estimating sideslip angle is not only critical for the trajectory tracking control of AVs but also essential for vehicle navigation, which is vital for communication among CAVs. Therefore, accurate sideslip angle estimation is indispensable for both AVs and CAVs.

III. TRAJECTORY TRACKING CONTROL OF AVS

Vehicle trajectory tracking control is a crucial and fundamental component of AVs, enabling the computation of direct optimal commands like steering angles, throttle opening, and braking pedal to replace and assist the driver [129, 130]. Within the control module, it can ensure that the vehicle accurately follows a predefined path while avoiding obstacles and maintaining safe driving conditions [131]. Accordingly, the feedback or self-learning tracking control algorithms at the microscopic level are designed to adjust the vehicle's speed, acceleration, and steering to follow the desired trajectory. Typically, the algorithms take into account the vehicle's dynamics and kinematics to determine the optimal control inputs. These inputs are adjusted in real-time to account for any uncertainties or disturbances in the vehicle's environment, such as changes in road conditions, traffic flow, or the presence of obstacles.

Nowadays, numerous institutions, automotive manufacturers, and component suppliers in the AV field have devoted significant attention to developing tracking control algorithms [14, 132–135]. These algorithms are primarily divided into the following three kinds: 1) feedback control without prediction (e.g., proportional-integral-derivative (PID), linear quadratic regulator (LQR), and sliding mode control (SMC) [136–138]); 2) feedback control with the prediction (e.g., model predictive control (MPC) [139–142]); and 3) learning-based control, such as the deep reinforcement learning. The overall framework is illustrated in Fig. 5.

A. Vehicle mathematical model

The trajectory tracking control of AVs involves the use of various vehicle models, which are typically classified into three types: VKB model [129, 143], bicycle VDB model [144–146], and four-wheel VDB model [102, 144, 147]. Different models have their own applicable scenarios [148].

The VKB model is widely used as it could establish a relationship between the lateral/heading error and the steering angle directly. In general, the pure pursuit (PP) method is employed to calculate the lateral error from a fixed look-ahead distance on the desired path ahead of the vehicle. Typically, the reference point on the vehicle for this method is the center of the rear axle. The corresponding model is as follows:

$$\begin{aligned}\delta_f &= \arctan\left(\frac{2L\sin\alpha}{L_d}\right) \\ \kappa &= \frac{2\sin\alpha}{L_d} \\ \sin\alpha &= \frac{e}{L_d}\end{aligned}\quad (9)$$

where δ_f is the steering angle; L is the wheelbase; α is the angle between the vehicle's body heading and the look-ahead direction; κ is the curvature; e is the lateral error; L_d is the look-ahead distance, which is a function of the vehicle velocity.

The equation above takes into account only the lateral error while disregarding the heading error, which is also crucial for ensuring optimal trajectory tracking performance. In order to address this issue, the Stanley method is developed, which considers both lateral and heading errors simultaneously. More specifically, the lateral error is defined as the distance between the closest point on the path and the front axle of the vehicle. The steering angle can be obtained using the following formula:

$$\delta_f = \arctan\left(\frac{ke}{1 + v(t)}\right) + \delta_h \quad (10)$$

where e is the lateral error between the closest point on the path with the front axle of the vehicle; $v(t)$ is the vehicle velocity; k is the feedback gain; δ_h is the heading error. Compared with the PP method, more parameters need to be fine-tuned to ensure convinced performance.

Although the VKB model is widely used in vehicle trajectory tracking control, it ignores errors like tire slip which is pretty important for critical driving conditions. In order to reduce the above errors, many scholars also construct vehicle trajectory tracking control algorithms with the VDB model. Here is an example of a widely used bicycle VDB model. Its formula is as follows:

$$\begin{aligned}\ddot{x} &= \ddot{y}\dot{\theta} + a_x \\ \ddot{y} &= -\dot{\varphi}^2 + \frac{2}{m}(F_{x1}\cos\delta_f + F_{x2}) \\ \ddot{\varphi} &= \frac{2}{L_f}(l_1 F_{x1} - l_2 F_{x2})\end{aligned}\quad (11)$$

In order to further improve the accuracy of the vehicle model, some scholars also adopt the four-wheel VDB model [149]. However, the improvement in accuracy will also lead to an increase in model complexity and computation. For this reason, the selection of the vehicle mathematical model ultimately depends on several factors, including driving scenarios, accuracy, and complexity.

B. Feedback control algorithms without prediction

Feedback control algorithms without prediction typically employ explicit control theories, such as PID, LQR, and SMC methods [150]. These techniques offer robust trajectory tracking performance under most driving conditions.

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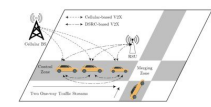


Fig. 12. Illustration of the collaborative driving scenario at intersections [306].

to improved fuel efficiency and reduced emissions.

To achieve collaborative merging, Ding et al. present a rule-based adjusting algorithm for coordinating the merging of two strings of vehicles at highway on-ramps efficiently and safely in the longitudinal direction and evaluates its performance through simulation-based case studies under both balanced and unbalanced scenarios [303]. Moreover, Liao et al. propose a collaborative ramp merging system for CAVs using a digital twin approach based on vehicle-to-cloud communication that provides advisory information to improve safety and environmental sustainability during merging with an acceptable communication delay, as demonstrated in a real-world field implementation in Riverside, California [304].

In order to reduce the complexity of problem-solving, Jing et al. present a hierarchical and decentralized collaborative coordination framework for the integrated longitudinal and lateral control of CAVs approaching on-ramps, which optimizes fuel consumption and passenger comfort, reduces fuel consumption, and improves traffic efficiency compared to baseline [305]. In addition, game theory has been widely used by researchers to model the collaborative control process. For example, Chen et al. propose a collaborative merging strategy for CAVs based on collaborative game theory and optimal control to improve traffic efficiency, reduce fuel consumption and enhance driving comfort through the construction of an economic payoff function that determines the merging sequence and coordinated merging trajectory [306].

For the scenario of mixed traffic, Sun et al. study collaborative control method for CAVs and conventional human-driven vehicles by developing a bi-level optimization program that guarantees system-efficient solutions and results in smoother ramp merging with an increase in traffic throughput of 10.15% [307]. Karimi et al. outlines a hierarchical control framework for merging areas in mixed traffic composed of CAVs and human-driven vehicles, with a focus on the lower level control algorithm that establishes a set of collaborative CAV trajectory optimization algorithms for different merging scenarios through the use of model predictive control [308].

b) Signalized intersections: In signalized intersections, collaborative control technology can be used to enable vehicles to communicate with each other and the infrastructure to optimize traffic flow and reduce the likelihood of accidents [310]. By enabling vehicles to anticipate and respond to traffic signals, collaborative control technology can help to reduce wait times and congestion, improve the safety of

turning movements, and enable more efficient use of intersection capacity [311–314]. For example, Wang et al. propose a collaborative eco-driving system for signalized corridors that uses a rule transition protocol for CAVs, leading to a reduction in energy consumption and pollutant emissions as CAV penetration rate increases [313]. Liu et al. present an approach that optimizes traffic signals and vehicle platooning at intersections through mixed integer linear programming, resulting in improved performance in terms of travel delay, throughput, fuel consumption, and emission compared to other approaches, which is verified by simulation [277].

c) Unsignalized intersections: In unsignalized or uncontrolled intersections, collaborative control technology can help to improve safety and efficiency by enabling vehicles to communicate with each other and coordinate their movements [315]. This can help to reduce the likelihood of accidents caused by miscommunication or failure to yield, and enable smoother and more efficient traffic flow through the intersection [316].

For example, in order to coordinate the motion of vehicles at unsignalized intersections, Deng et al. propose a conflict duration graph-based coordination framework to resolve collisions and improve traffic capacity according to heuristic rules at signal-free intersections [317]. Deng et al. further extend the research on collaborative control by proposing a bi-level optimization method to coordinate the merging of CAVs at unsignalized intersections, aiming to achieve an optimal traffic schedule with dynamically-feasible and energy-efficient trajectories that improve space utilization and prevent spillbacks [309]. In addition, Wang et al. present a digital twin framework for a collaborative control system at non-signalized intersections, which incorporates an enhanced FIFO slot reservation algorithm, a consensus motion control algorithm, a model-based motion estimation algorithm, and an augmented reality HMI [318].

Overall, this section discusses collaborative control techniques for CAVs, which are crucial for realizing their potential to improve transportation systems [319–321]. It covers enabling technologies, critical components, collaborative control methodologies, and potential applications. The enabling technologies provide the communication infrastructure for CAVs to share information and cooperate. Critical components such as longitudinal dynamics models, information flow topology, communication quality, and stability are necessary for collaborative control. Collaborative control methodologies include centralized, decentralized, or mixed control mode, and utilize game-theoretic, optimization-based, or reinforcement learning-based methods. The potential applications of collaborative control include platooning, lane changing, intersection crossing, and more, which can enhance traffic efficiency, reduce emissions, and improve safety in one-, two-, or three-dimensional collaboration scenarios.

V. CONCLUSION AND FUTURE WORKS

In this survey, we have provided a comprehensive review of vehicle control for autonomous vehicles and connected and automated vehicles. We began with vehicle state estimation,

结论或局限页 / Conclusion or limitations page

p.18 · full_page

这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。

This snapshot anchors the study note to the original PDF page so you can read the explanation alongside the source layout.

与当前课题的连接：Useful context citation for why trajectory tracking, collaborative control, safety, comfort, and energy belong together.

章节精读 / Section-by-Section Deep Reading

p.1 · motivation

I. INTRODUCTION

章节目标：建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Learning goal: Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 CAV control survey：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.3 · bridge

II. VEHICLE SIDESLIP ANGLE ESTIMATION

章节目标：承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.8 · method

III. TRAJECTORY TRACKING CONTROL OF AVS

章节目标：解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Learning goal: Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 作为综述，它按估计、跟踪、路径规划、协同控制等层次整理方法。对本项目最有价值的是把 comfort、safety、energy、efficiency 放入同一个 CAV 控制目标谱系。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.13 · method

IV. COLLABORATIVE CONTROL OF CAVS

章节目标：解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Learning goal: Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 作为综述，它按估计、跟踪、路径规划、协同控制等层次整理方法。对本项目最有价值的是把 comfort、safety、energy、efficiency 放入同一个 CAV 控制目标谱系。

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审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.18 · critique

p.20 · bridge

V. CONCLUSION AND FUTURE WORKS

章节目标：提炼作者承认的边界，并把它转化为博士课题的可拓展问题。

Learning goal: Convert acknowledged boundaries into research opportunities.

关键论点 / Key argument: 结论不只是总结结果，更是观察作者没有解决什么的窗口。

技术焦点 / Technical focus: 把 future work 与前文假设逐一对应，寻找可发表的补强点。

审稿追问 / Reviewer question: 如果我是审稿人，我会要求补哪一个实验、定理或消融？

X. Wang, “Federated vehicular transformers and their

章节目标：承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.23 · bridge

X. Cao, P. Zhang, Y . Cao, L. Qi, et al. , “Multi–

章节目标：承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

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审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.24 · bridge

X. Cheng, “Metasaug: Meta semantic augmentation for

章节目标：承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

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审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段

先把作者的痛点翻译成一句博士问题：CAV 控制从状态估计、轨迹跟踪到协同控制有哪些主线方法，它们如何支撑交叉口协同研究？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段

把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：文献覆盖广但不可避免存在选择偏差。；不同控制任务和应用场景的指标不可直接比较。；综述结论需要回到具体论文验证。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段

围绕核心公式“通用车辆控制闭环 / Generic vehicle-control loop”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段

把算法读成一条数据流：输入交通状态，执行 Systematic survey from vehicle state estimation and trajectory tracking to collaborative CAV control applications., 输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段

不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Literature survey rather than original experiments.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段

最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Useful context citation for why trajectory tracking, collaborative control, safety, comfort, and energy belong together.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

技术路线图与贡献量评估 / Technical Route & Contribution Matrix

1. 输入 Input 车辆状态、路径/车道、交通需求、信号或协同信息。	2. 建模 Modeling 通用车辆控制闭环 / Generic vehicle-control loop	3. 核心求解 Core solver Systematic survey from vehicle state estimation and trajectory tracking to collaborative CAV control applications.
4. 安全接口 Safety interface Reviews vehicle-control approaches relevant to safety and collaborative control.	5. 平滑与效率 Smoothing and efficiency Frames comfort, energy saving, and transportation efficiency as central CAV control goals.	6. 验证 Validation Literature survey rather than original experiments.

贡献量评估 / Contribution Strength Matrix

Dimension / 维度	Score / 评分	Comment / 评语
理论贡献 / Theoretical contribution	3/5	通用车辆控制闭环 / Generic vehicle-control loop
算法贡献 / Algorithmic contribution	4/5	Systematic survey from vehicle state estimation and trajectory tracking to collaborative CAV control applications.
实验贡献 / Experimental contribution	2/5	Literature survey rather than original experiments.
工程贡献 / Engineering contribution	4/5	Reviews vehicle-control approaches relevant to safety and collaborative control.
博士课题价值 / PhD relevance	4/5	Useful context citation for why trajectory tracking, collaborative control, safety, comfort, and energy belong together.

2. 理论方法论深度解构 / Deep Dive into Methodology & Theoretical Framework

问题数学建模 / Mathematical Formulation

通用车辆控制闭环 / Generic vehicle-control loop

$$\begin{aligned} \dot{x} &= f(x, u, w), \quad y = g(x), \quad u = \kappa(\hat{x}, r), \quad e = r - y \end{aligned}$$

Symbol / 符号	含义	Meaning	Role / 建模作用
x	车辆状态	vehicle state	控制系统内部变量
u	控制输入	control input	转向、加速度或制动力
w	外部扰动	external disturbance	道路、交通和模型误差
$\hat{x}$	估计状态	estimated state	状态估计模块输出
r	参考轨迹	reference trajectory	轨迹规划/调度层给定目标

核心算法架构 / Core Algorithm Layout

作为综述，它按估计、跟踪、路径规划、协同控制等层次整理方法。对本项目最有价值的是把 comfort、safety、energy、efficiency 放入同一个 CAV 控制目标谱系。

As a survey, it organizes estimation, tracking, planning, and cooperative-control techniques, framing safety, comfort, energy, and efficiency as connected CAV objectives.

收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

综述不提供单个算法的收敛证明，但能帮助你定位不同方法的理论依据，如鲁棒控制、MPC、学习控制和协同控制。

It does not certify one algorithm, but it maps the theoretical families used to justify robust control, MPC, learning control, and cooperative control.

潜在假设与边界条件 / Assumptions & Boundary Conditions

- 文献覆盖广但不可避免存在选择偏差。
- 不同控制任务和应用场景的指标不可直接比较。
- 综述结论需要回到具体论文验证。

3. 实验验证与结果评判 / Experimental Validation & Result Evaluation

基准对比与环境 / Baselines & Environment

不适用原始实验；基准是分类完整性、代表性文献覆盖和趋势判断。
Literature survey rather than original experiments.

核心指标 / Metrics

综述质量、分类清晰度、引用覆盖、应用关联度和未来方向启发。

消融实验 / Ablation

综述没有消融；阅读时应做你自己的 taxonomy cross-check：是否遗漏 intersection scheduling、formal safety、mixed traffic。

博士阅读提醒 / Reading Reminder

阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 20 figure references and 0 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
控制框架/轨迹曲线图	control-framework or trajectory-profile plots	关注约束激活位置、速度/加速度曲线和安全裕度是否平滑。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

4. 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

论文局限性 / Limitations & Weaknesses

- 范围太广，交叉口轨迹平滑不是焦点。
- 综述时效性随新 RL/CBF/GNN 文献快速下降。
- 缺少统一实验平台比较。

博士课题拓展点 / Research Extensions for PhD

- 把综述分类裁剪成你的 thesis taxonomy: scheduler、smoother、safety filter、evaluation。
- 建立一张 CAV 控制方法与 JSSP/RL/AIM 的映射表。
- 用它支持引言中 safety-comfort-energy-efficiency 多目标论证。

如何服务当前课题 / Use in This Project

Useful context citation for why trajectory tracking, collaborative control, safety, comfort, and energy belong together.

5. 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数探索 / Parameter Exploration

方法成熟度 / Method maturity:

0.65

成熟度越高可验证性越强，但探索性和性能上限可能较低。

Higher maturity improves certifiability but may reduce exploratory performance.

HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar：题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 `input[type=range]` + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

以下锚点来自本地 PDF 抽取，用于回到原文复核；笔记不保存论文全文。

Page	Extracted heading
p.1	I. INTRODUCTION
p.3	II. V EHICLE SIDESLIP ANGLE ESTIMATION
p.18	V. CONCLUSION AND FUTURE WORKS

p.1 · I. INTRODUCTION

定位问题背景、研究缺口和为什么现有保守/非协同方法不足。
Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.3 · II. V EHICLE SIDESLIP ANGLE ESTIMATION

作为局部论证段落阅读，关注它如何服务主问题。
Read as a local argument and ask how it supports the main question.

p.8 · III. T RAJECTORY TRACKING CONTROL OF AVS

给出算法流程和求解机制，应重点追踪输入、输出和安全接口。
Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.13 · IV. C OLLABORATIVE CONTROL OF CAVS

给出算法流程和求解机制，应重点追踪输入、输出和安全接口。
Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.18 · V. CONCLUSION AND FUTURE WORKS

总结贡献和未来方向，也常暴露作者承认的边界条件。
Summarizes contributions and often reveals author-recognized boundaries.

p.20 · X. Wang, “Federated vehicular transformers and their

作为局部论证段落阅读，关注它如何服务主问题。
Read as a local argument and ask how it supports the main question.

p.23 · X. Cao, P. Zhang, Y . Cao, L. Qi, et al. , “Multi-

作为局部论证段落阅读，关注它如何服务主问题。

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